Code No: 118BJ **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July - 2021
EMBEDDED SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks

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1. Explain the five levels of abstraction from top level to bottom level in the design process of an embedded system. [15]

2. With a neat internal diagrams, describe all the input/output ports of 8051 μC. [15]

3. With one example for each, explain the following instructions by giving syntaxes:
   a) Data transfer instructions b) Arithmetic instructions. [8+7]

4. Draw the block schematic of DAC 0808 interfaced to 8051 at port P1 and write an 8051 program to generate square wave as well as triangular wave. [15]

5. Discuss the following with respect to RTOS:
   a) Semaphores b) Message Queues. [8+7]

6.a) Explain different debugging techniques.
   b) List out the features of ARM architecture. [8+7]

7. What is the criterion used to select the time quantum in case of round-robin scheduling algorithm? Explain it with a suitable example. [15]

8.a) Explain the basic concepts of I2C bus protocol.
   b) Explain the general purpose registers in SHARC programming model. [8+7]

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